

PROJECT ASSIGNMENT 2

Group10-PA2 User Research

FIFA.com and Chess.com

Course	CSC13112 - UI/UX Design, FIT-HCMUS
Group	Group10
Team	Le Minh (21127645); Nguyen Vu Bach (21127224); Pham Nguyen Gia Bao (20127119); Trang Minh Nhut (22127318)

Contents

1. Research scope, methods, and validity boundary	— 3
2. Tasks and protocol	— 4
2.1 FIFA task set	— 4
2.2 Chess task set	— 4
3. Simulated participants	— 5
4. Three-day simulated session plan	— 8
5. Representative Sessions and Patterns	— 10
5.1 Pattern Summary	— 10
5.2 Result Summary	— 10
6. Simulated task results	— 11
7. Personas	— 12
7.1 FIFA trust-sensitive ticket planner	— 12
7.2 FIFA returning tournament follower	— 13
7.3 Chess first-time beginner	— 13
7.4 Chess returning learner who forgets analysis terms	— 14
8. Representative Screenshot Evidence	— 16
9. Findings	— 27
9.1 Findings Summary	— 28
10. Evidence boundary and citations	— 28
11. Revised Work Models	— 29
Work Model Coverage Checklist	— 35
12. Practical Difficulties and Lessons Learned	— 36
Appendix A - Twelve Detailed Simulated Note Sheets	— 37

A.1 Note sheet F-SIM-01	— 37
A.2 Note sheet F-SIM-02	— 38
A.3 Note sheet F-SIM-03	— 39
A.4 Note sheet F-SIM-04	— 40
A.5 Note sheet F-SIM-05	— 41
A.6 Note sheet F-SIM-06	— 42
A.7 Note sheet C-SIM-01	— 43
A.8 Note sheet C-SIM-02	— 44
A.9 Note sheet C-SIM-03	— 45
A.10 Note sheet C-SIM-04	— 46
A.11 Note sheet C-SIM-05	— 47
A.12 Note sheet C-SIM-06	— 48

1. Research scope, methods, and validity boundary

Simulated-study disclosure: All participant profiles, sessions, quotations, observations, task outcomes, and ratings in this report are scenario-based synthetic evidence created to test design hypotheses; they are not presented as research with real users.

Method	What it supports	Boundary	Source label
Source analysis	Interface inventory, documented states, PA1 continuity	A file or screen does not prove behavior	Screen evidence / PA1 inherited finding
Try-it-yourself walkthrough	Task path, transition, recovery, and handoff inspection	Shows the path available to the evaluator, not population performance	Screen evidence / Design inference
Simulated user sessions	Scenario coverage, test cases,	Does not represent real-user	Simulated study / PA3

Method	What it supports	Boundary	Source label
	hypotheses, and comparison logic	measurement	validation target

Course source: Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 - Fundamental Concepts - Usability Dimensions_2.pdf, pp. 62-67, 74-76, 83-84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36; LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

PA1 inherited finding: FIFA separates useful tournament content from ticket decision confidence; Chess.com provides strong play and learning functions but exposes analysis choices with high recall demand.

Design inference: Visibility of system status, recognition over recall, progressive disclosure, user control, and error prevention organize the PA2 questions.

2. Tasks and protocol

2.1 FIFA task set

1. Find ticket entry.
2. Compare tournament status.
3. Decide buy, wait, register interest, resale, or hospitality.
4. Identify destination before leaving FIFA.com.
5. Resume on mobile.
6. Recover from missing or stale status.

2.2 Chess task set

1. Choose a beginner route.
2. Continue a lesson.
3. Solve a puzzle.
4. Open analysis or review.
5. Identify one important mistake.
6. Try a better move.
7. Read a plain-language explanation.
8. Continue to practice.
9. Reveal advanced detail on demand.

Simulated research session protocol: Each 25-35 minute record uses 5 minutes of introduction, 15-20 minutes of task work, and 5 minutes of follow-up. Notes are structured; no audio or video is recorded.

3. Simulated participants

ID	Role	Age	Device	Domain / application experience	Frequency	Motivation and context	Constraints / accessibility	Tasks
F-SIM-01	Casual football fan	18-24	Android phone	Follows major finals; Occasional FIFA.com visitor	Monthly in season	Find ticket entry without planning expertise; Commute; one-handed use	Limited time and data; Large tap targets	F1-F6
F-SIM-02	Casual football fan	25-34	Windows laptop + phone	Watches selected national-team matches; Uses Match Centre occasionally	Several times per tournament	Compare match and ticket status; Evening planning at home	Avoids creating accounts early; High contrast	F1-F6
F-SIM-03	Tournament follower	25-34	iPhone	Tracks brackets and schedules; Returning FIFA.com visitor	Weekly in tournament periods	Coordinate a group trip; Mobile-first, shared planning	Needs fast resumption; Text zoom	F1-F6
F-	Tournament	35-	MacBook	Follows multiple	Weekly	Connect schedule	Many open tabs;	F1-

ID	Role	Age	Device	Domain / application experience	Frequency	Motivation and context	Constraints / accessibility	Tasks
SIM-04	t follower	44	+ Android phone	competitions; Regular Match Centre user		context to ticket action; Desk-to-mobile transition	Keyboard access on laptop	F6
F-SIM-05	Trust-sensitive ticket planner	35-44	Windows laptop	Has bought event tickets online; Knows FIFA.com but not partner boundaries	At key sale windows	Avoid unofficial or stale destinations; Family purchase planning	Needs provenance before leaving site; Clear status text, not color alone	F1-F6
F-SIM-06	Trust-sensitive ticket planner	45-54	Android phone	Plans international tournament travel; Low-frequency FIFA.com visitor	A few times per year	Know whether to buy, wait, or register interest; Mobile on variable network	Low confidence after handoff; Plain language and persistent focus	F1-F6
C-SIM-01	First-time beginner	18-24	Android phone	Knows piece movement; First Chess.com week	Daily trial week	Choose a safe beginner route; Short study breaks	Unfamiliar analysis terms; Plain language	C1-C9
C-SIM-02	First-time beginner	25-34	Windows laptop	Played fewer than ten games; New account	Twice weekly	Understand one mistake; Home learning session	Fears breaking the position; Keyboard-only path	C1-C9

ID	Role	Age	Device	Domain / application experience	Frequency	Motivation and context	Constraints / accessibility	Tasks
C-SIM-03	Returning learner	18-24	iPhone	Completed basic lessons; Returns after four months	Weekly when active	Resume without relearning navigation; Mobile evening practice	Forgets feature locations; Text zoom	C1-C9
C-SIM-04	Returning learner	35-44	MacBook	Knows opening principles; Intermittent lesson and puzzle use	Monthly	Reconnect review to practice; Desktop at home	Forgets analysis vocabulary; Reduced motion	C1-C9
C-SIM-05	Intermediate player with low analysis use	25-34	Windows laptop	Club-level casual; Plays games and puzzles	Weekly	Review without engine overload; After-game desktop session	Skips dense panels; Keyboard shortcuts with labels	C1-C9
C-SIM-06	Intermediate player with low analysis use	35-44	Android tablet	Understands tactics; Rarely opens analysis	Several times per month	Find one actionable improvement; Tablet practice session	Terminology slows scanning; Color-independent indicators	C1-C9

Scenario-based synthetic evidence: IDs are fictional study roles; no names identify participants.

4. Three-day simulated session plan

Sessi on ID	Date	Dura tion	Partici pant	Device	Tasks	Facilitator	Note taker	Recording method
F- SES-0 1	16 Jul 2026	28 min	F- SIM-0 1	Android phone	F1, F2, F4, F5	Nguyen Vu Bach	Trang Minh Nhut	Structured notes only; no audio/video
F- SES-0 2	16 Jul 2026	31 min	F- SIM-0 2	Windows laptop + phone	F1-F6	Nguyen Vu Bach	Le Minh	Structured notes only; no audio/video
C- SES-0 1	16 Jul 2026	29 min	C- SIM-0 1	Android phone	C1-C3, C7-C9	Pham Nguyen Gia Bao	Trang Minh Nhut	Structured notes only; no audio/video
C- SES-0 2	16 Jul 2026	32 min	C- SIM-0 2	Windows laptop	C1-C9	Pham Nguyen Gia Bao	Le Minh	Structured notes only; no audio/video
F- SES-0 3	17 Jul 2026	30 min	F- SIM-0 3	iPhone	F1-F6	Nguyen Vu Bach	Trang Minh Nhut	Structured notes only; no audio/video
F- SES-0 4	17 Jul 2026	34 min	F- SIM-0 4	MacBook + Android phone	F1-F6	Nguyen Vu Bach	Le Minh	Structured notes only; no audio/video
C- SES-0	17 Jul 2026	27 min	C- SIM-0	iPhone	C1-C3, C5-C9	Pham Nguyen	Trang Minh Nhut	Structured notes only; no

Sessi on ID	Date	Dura tion	Partici pant	Device	Tasks	Facilitator	Note taker	Recording method
3			3			Gia Bao		audio/video
C- SES-0 4	17 Jul 2026	35 min	C- SIM-0 4	MacBook	C1-C9	Pham Nguyen Gia Bao	Le Minh	Structured notes only; no audio/video
F- SES-0 5	18 Jul 2026	33 min	F- SIM-0 5	Windows laptop	F1-F6	Nguyen Vu Bach	Trang Minh Nhut	Structured notes only; no audio/video
F- SES-0 6	18 Jul 2026	35 min	F- SIM-0 6	Android phone	F1-F6	Nguyen Vu Bach	Le Minh	Structured notes only; no audio/video
C- SES-0 5	18 Jul 2026	30 min	C- SIM-0 5	Windows laptop	C1-C9	Pham Nguyen Gia Bao	Trang Minh Nhut	Structured notes only; no audio/video
C- SES-0 6	18 Jul 2026	34 min	C- SIM-0 6	Android tablet	C1-C9	Pham Nguyen Gia Bao	Le Minh	Structured notes only; no audio/video

Simulated research session: This is a simulated session record used for scenario coverage and planning, not an attendance or research claim.

5. Representative Sessions and Patterns

Six representative sessions keep the main report concise. The remaining detailed note sheets are retained in Appendix A.

Session	Task focus	Observed pattern	Recovery or result
F-SES-01	Find ticket entry on mobile	The entry was found, but the selected tournament moved out of view.	Return to the tournament summary before choosing a route.
F-SES-04	Connect Match Centre to ticket action	Schedule context did not fully carry into ticket planning.	Re-select the event and verify the current ticket state.
F-SES-05	Compare ticket status and destination	Freshness and partner boundaries shaped confidence.	Stay on FIFA.com when the state or destination is unclear.
C-SES-01	Choose a beginner route	Several entry choices increased recall demand.	Return to the recent game and select Beginner Review.
C-SES-04	Continue from explanation to practice	The next relevant lesson or puzzle was not obvious.	Use one mistake-linked practice recommendation.
C-SES-06	Review a game on a smaller screen	Dense analysis competed with the learning priority.	Show one mistake first and disclose advanced depth later.

5.1 Pattern Summary

FIFA patterns center on ticket-state clarity, freshness, mobile context, and partner handoff trust. Chess patterns center on entry choice, beginner explanation, and a clear continuation into practice.

5.2 Result Summary

The simulated sessions support two PA3 hypotheses: a FIFA decision layer should keep state, freshness, route, and recovery together; a Chess beginner review should show one interpretable mistake and one next practice action before advanced analysis.

6. Simulated task results

Trace	Task completion	Hesitations	Wrong paths	Recoveries	Confidence	Clarity
F-SIM-01 / F-SES-01	3/4	3	1	1	3/5	3/5
F-SIM-02 / F-SES-02	5/6	2	1	1	3/5	4/5
F-SIM-03 / F-SES-03	4/6	3	1	1	3/5	3/5
F-SIM-04 / F-SES-04	4/6	3	0	1	2/5	3/5
F-SIM-05 / F-SES-05	4/6	4	1	1	2/5	3/5
F-SIM-06 / F-SES-06	3/6	4	2	2	2/5	2/5
C-SIM-01 / C-SES-01	5/6	3	2	1	3/5	3/5
C-SIM-02 / C-SES-02	7/9	4	2	1	3/5	3/5
C-SIM-03 / C-SES-03	6/8	3	2	2	3/5	3/5
C-SIM-04 / C-SES-04	7/9	3	1	1	3/5	4/5
C-SIM-05 / C-SES-05	7/9	3	2	1	3/5	3/5
C-SIM-06 / C-SES-06	8/9	2	1	1	4/5	4/5

Simulated task result: The FIFA status/handoff records contain lower confidence and more hesitation than schedule-oriented work. Chess analysis entry has more wrong paths than puzzles; guided lesson work is clearer than analysis entry. No row is perfect, and every number traces to

Trace	Task completion	Hesitations	Wrong paths	Recoveries	Confidence	Clarity
one note sheet.						

7. Personas

7.1 FIFA trust-sensitive ticket planner

Field	Record
Persona ID	UR-P-F1
Summary	FIFA trust-sensitive ticket planner
Goals	Compares state and official destination before departure
Behaviors	Checks labels, timestamp, domain, and return route
Context	Family purchase at a sale window
Domain / application experience	High ticket-domain experience; moderate FIFA.com experience
Device pattern	Laptop first; phone resume
Frustrations	Ambiguous status and unexplained handoff
Capabilities	Careful comparison and risk detection
Limitations	Will wait rather than accept uncertainty
Simulated quote	"I need the state and destination before I leave."
Scenario	Compare sale, resale, hospitality, then choose act or stay.
Evidence basis	F-SIM-05/06; F2-E09; PA1 inherited ticket drawback
Assumptions to	Whether destination preview improves calibrated

Field	Record
validate	trust

7.2 FIFA returning tournament follower

Field	Record
Persona ID	UR-P-F2
Summary	FIFA returning tournament follower
Goals	Reconnect schedule context to ticket action
Behaviors	Starts in Match Centre and expects tournament continuity
Context	Weekly tournament planning
Domain / application experience	High competition knowledge; regular FIFA.com use
Device pattern	Phone first; laptop comparison
Frustrations	Mobile stack and lost filters
Capabilities	Fast recognition of tournaments and schedules
Limitations	Does not remember ticket channel wording
Simulated quote	"Keep my tournament context when I switch devices."
Scenario	Follow a tournament, inspect status, resume on mobile.
Evidence basis	F-SIM-03/04; F2-E03/E04; PA1 continuity
Assumptions to validate	Whether persisted context reduces recovery work

7.3 Chess first-time beginner

Field	Record
Persona ID	UR-P-C1

Field	Record
Summary	Chess first-time beginner
Goals	Understand one important mistake and next action
Behaviors	Uses Learn and Puzzles; avoids engine terminology
Context	Short mobile learning session
Domain / application experience	Basic rules; first week on Chess.com
Device pattern	Phone; occasional laptop
Frustrations	Too many entry choices
Capabilities	Willing to try a move and read a short explanation
Limitations	Cannot interpret evaluation vocabulary
Simulated quote	"Show me what to learn first."
Scenario	Complete a lesson, puzzle, and beginner review.
Evidence basis	C-SIM-01/02; C2-E05-E10; PA1 learning strength
Assumptions to validate	Whether one preset improves entry selection

7.4 Chess returning learner who forgets analysis terms

Field	Record
Persona ID	UR-P-C2
Summary	Chess returning learner who forgets analysis terms
Goals	Resume review and practice after a gap

Field	Record
Behaviors	Recognizes recent game and lesson outcomes over feature names
Context	Intermittent desktop session
Domain / application experience	Moderate chess knowledge; low analysis use
Device pattern	Laptop; tablet backup
Frustrations	Recall-heavy labels and dense advanced panels
Capabilities	Can compare moves with plain explanations
Limitations	Forgets analysis terminology
Simulated quote	"I remember the game, not the analysis label."
Scenario	Open recent game, identify one mistake, continue practice.
Evidence basis	C-SIM-03/04/05; C2-E07-E10
Assumptions to validate	Whether outcome-oriented labels improve memorability

8. Representative Screenshot Evidence

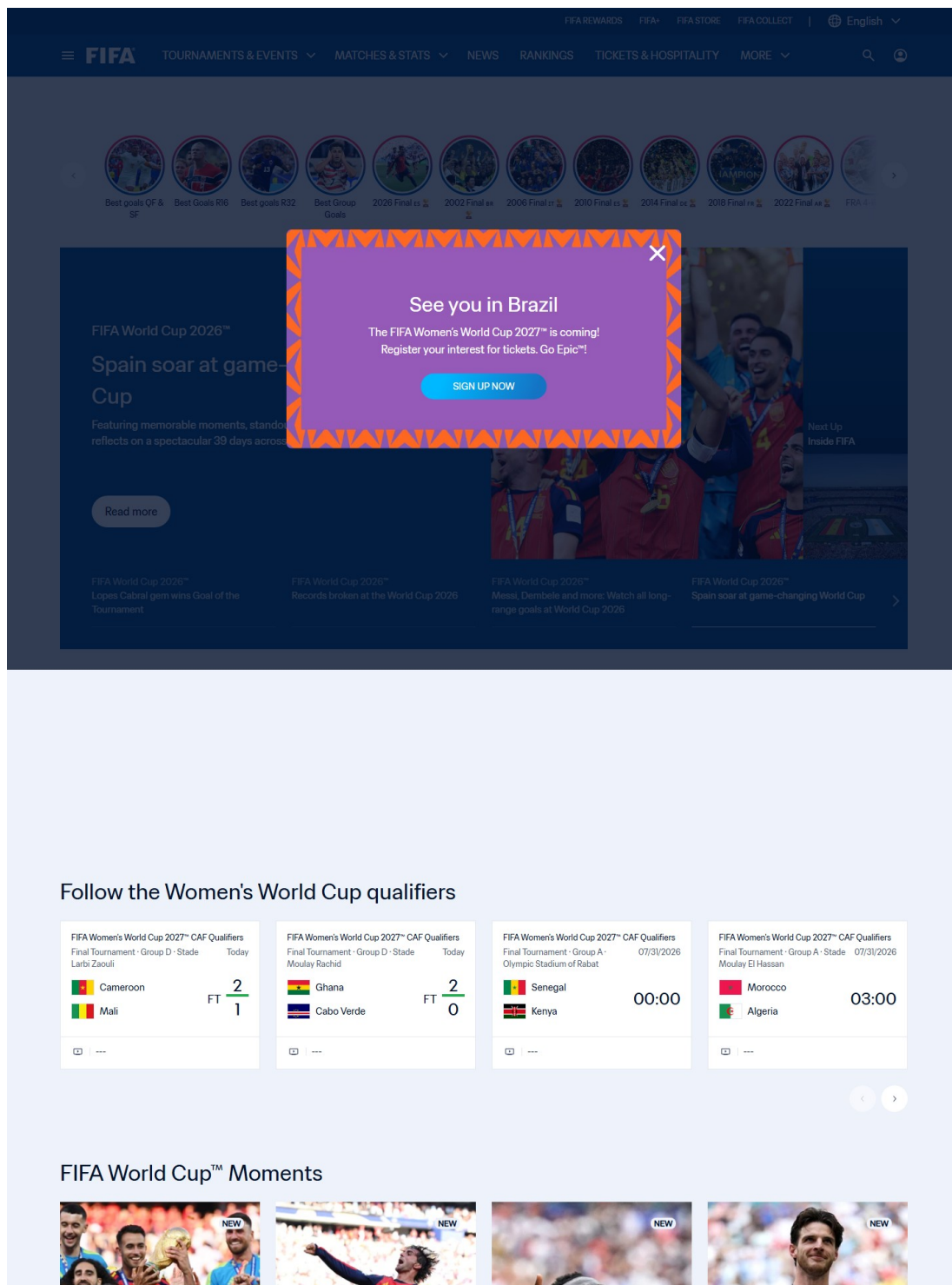


Figure UR-EV-03. State: FIFA homepage / entry. What is visible: Homepage hero, navigation, and entry choices. Why it matters: Establishes the starting context. Limitation: The capture does not prove which route a user chooses.

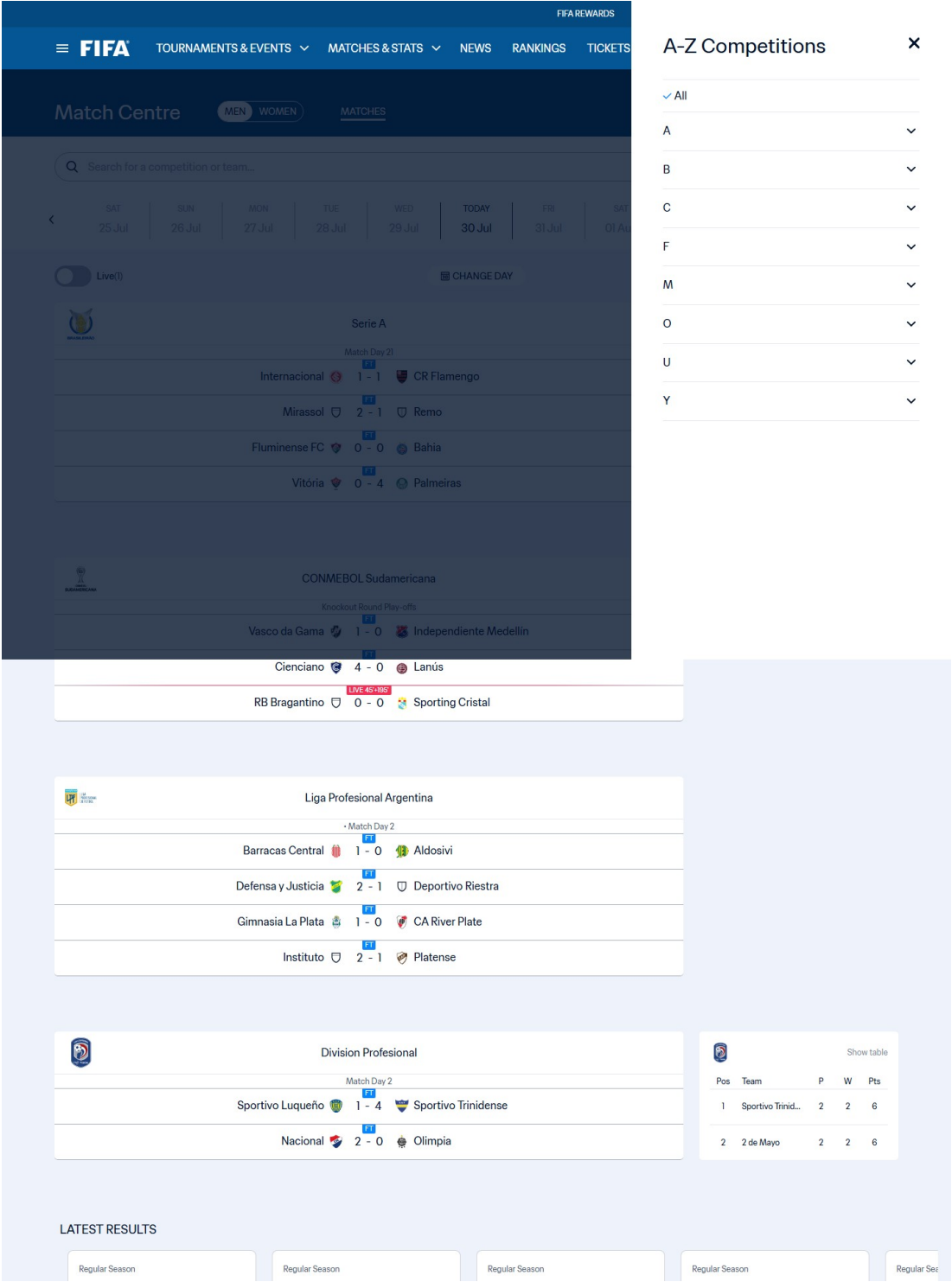


Figure UR-EV-04. State: FIFA Match Centre filters. What is visible: Date, competition, and match filtering controls. Why it matters: Shows the context users must preserve. Limitation: Live match availability changes over time.

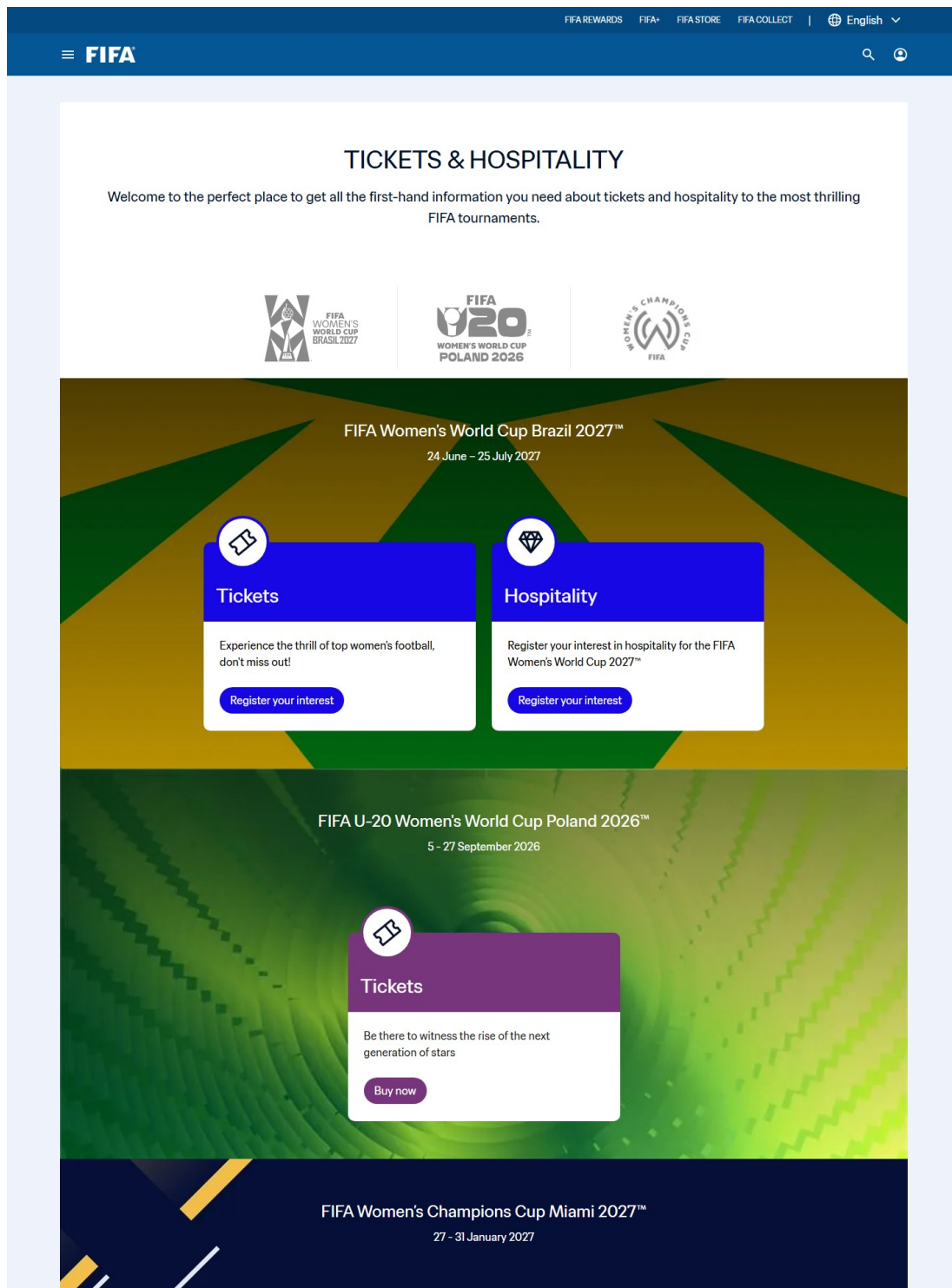


Figure UR-EV-05. State: Tickets and Hospitality landing. What is visible: Ticket and hospitality entry cards. Why it matters: Shows multiple planning routes. Limitation: It does not confirm current inventory.

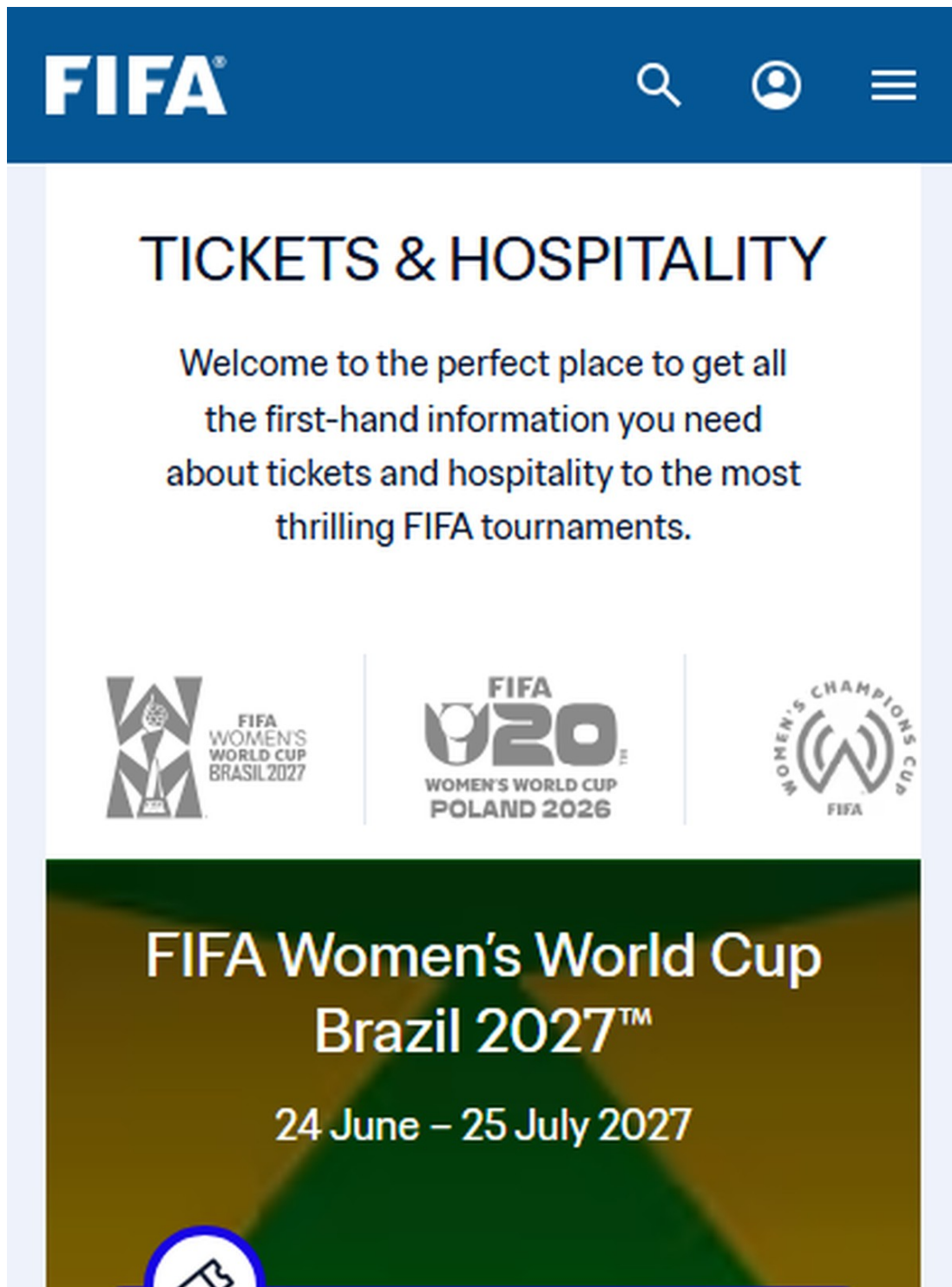


Figure UR-EV-06A. State: Mobile tickets - top segment. What is visible: The first readable ticket cards. **Why it matters:** Shows the start of the mobile decision surface. **Limitation:** This is only the top window of a long page.

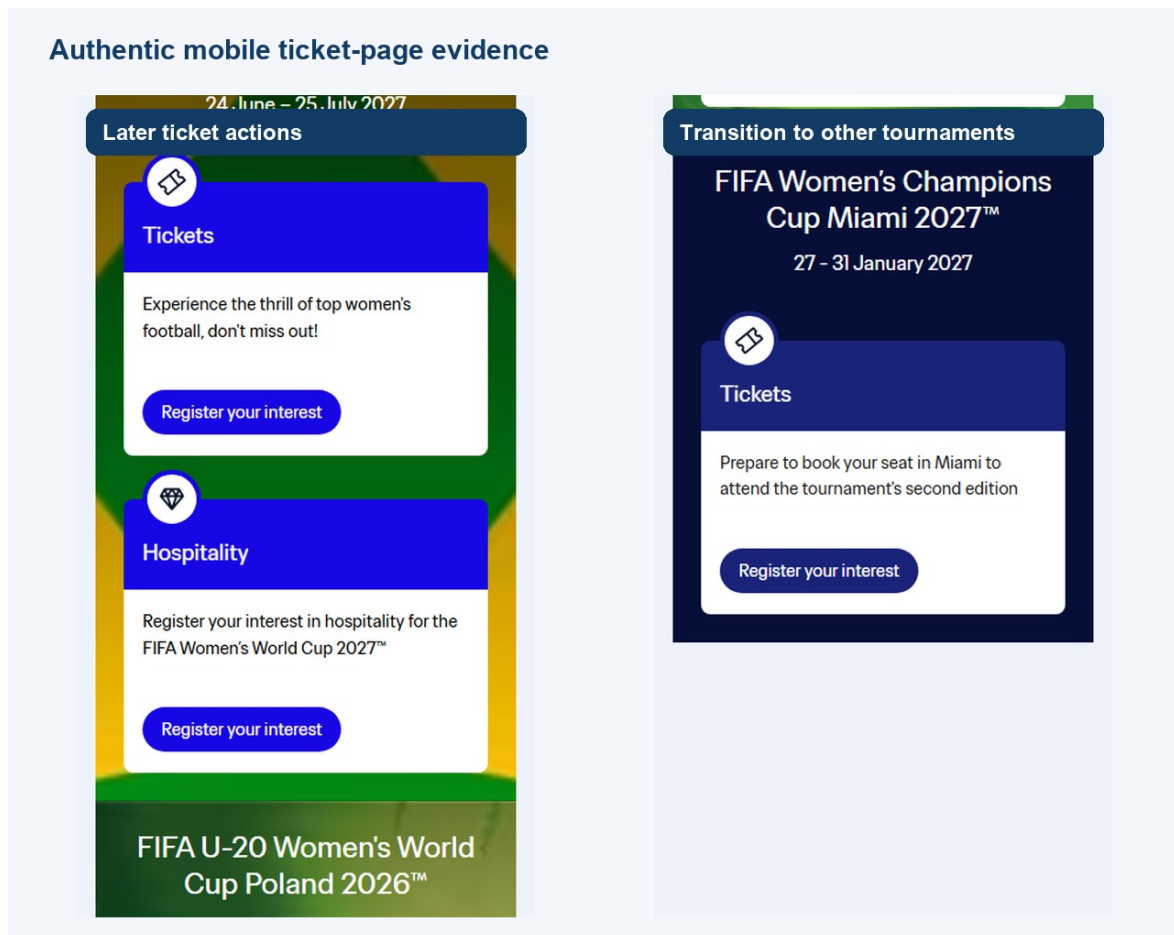


Figure UR-EV-06B. State: Mobile tickets - later action cards. Authentic crops show later ticket actions and the transition to other tournaments; they do not use footer-only content as action evidence.

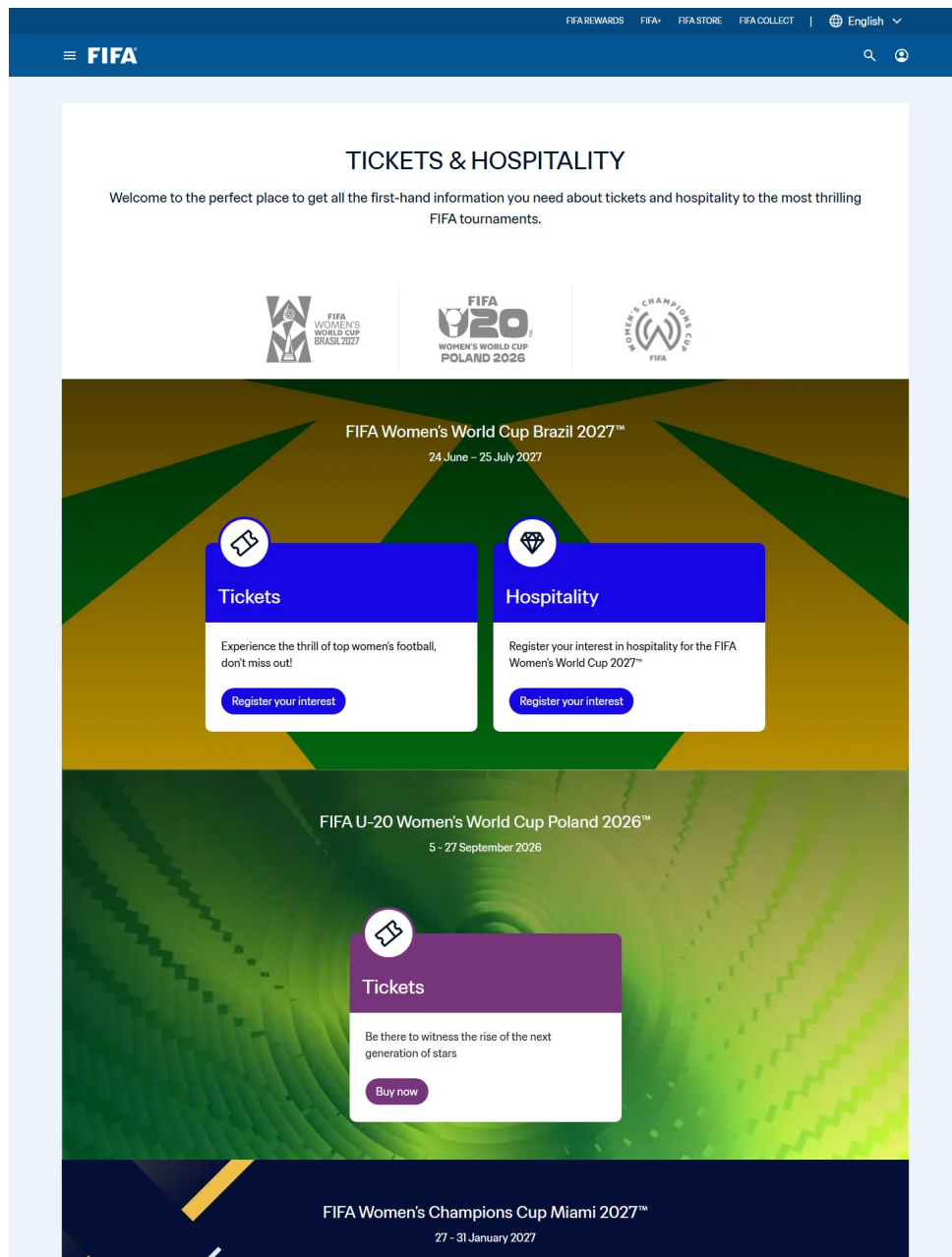


Figure UR-EV-07. State: Pre-handoff ticket route state. The captured redirect remained on FIFA Tickets & Hospitality, so this evidence documents the route before partner handoff and does not claim that a partner destination loaded.

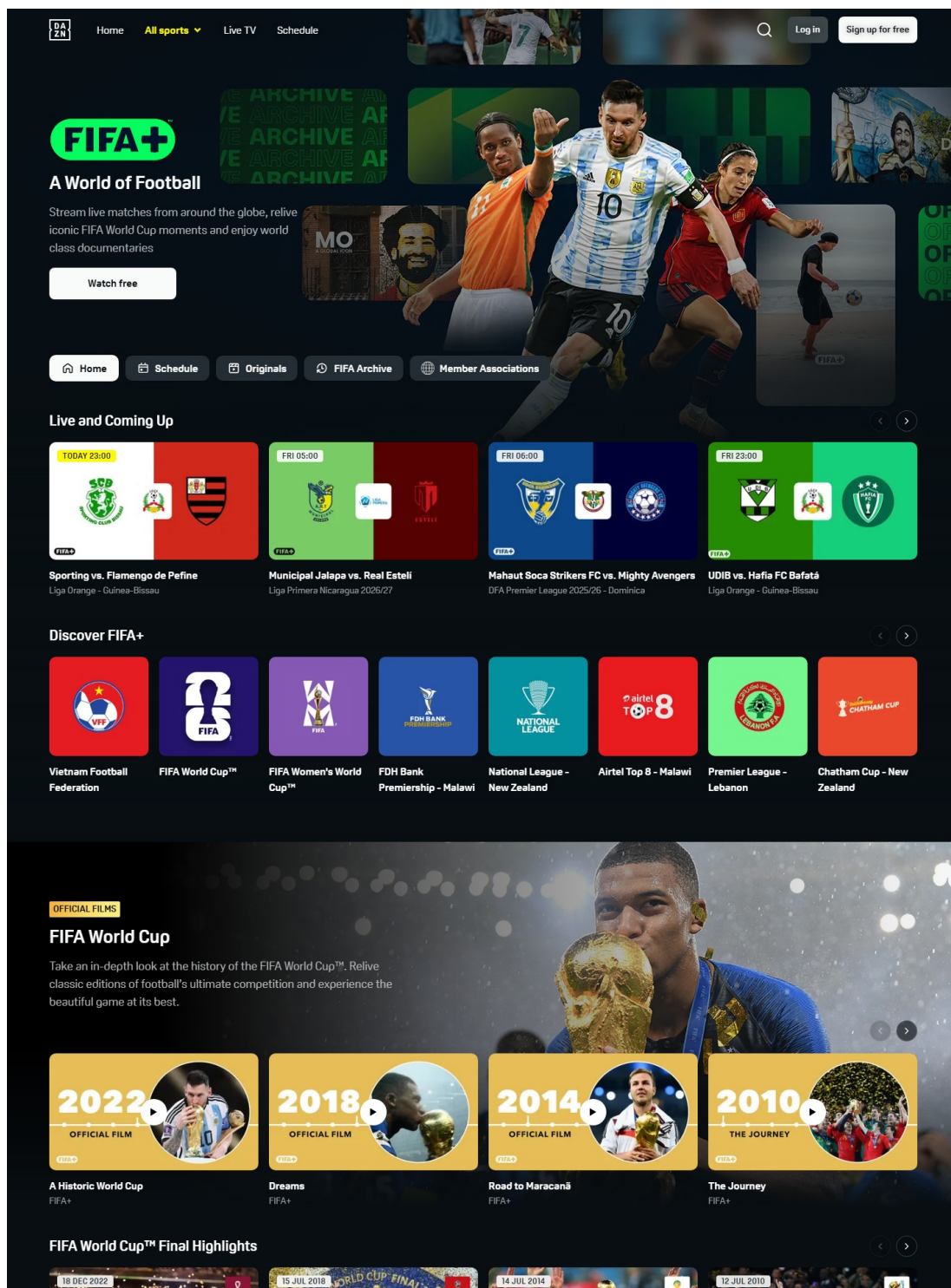


Figure UR-EV-08. State: FIFA+ destination. What is visible: External media destination and branding. Why it matters: Shows another handoff boundary. Limitation: Availability and account rules may vary.

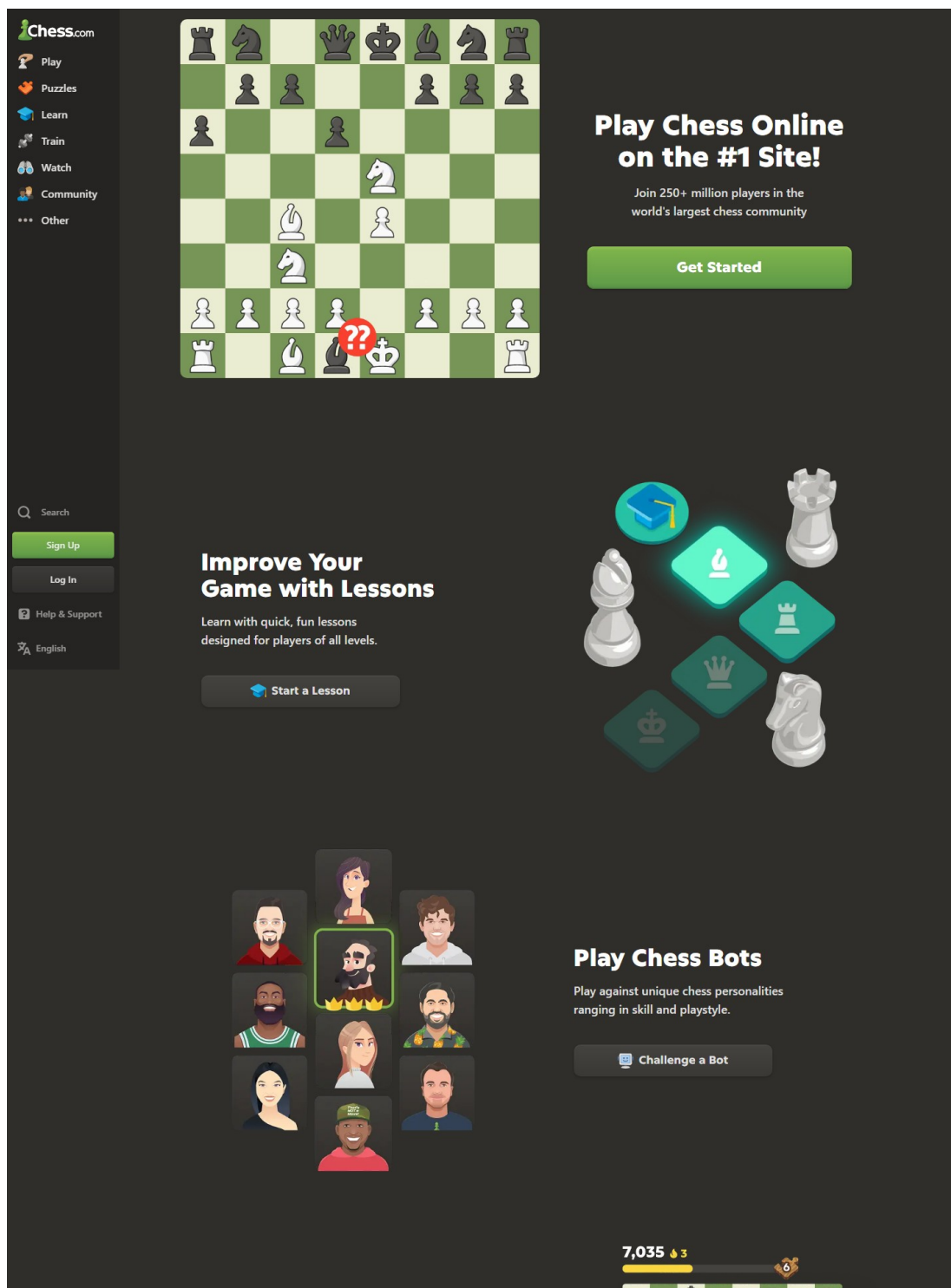


Figure UR-EV-09. State: Chess homepage. What is visible: Play and learning entry choices. Why it matters: Establishes the beginner starting point. Limitation: The capture does not measure choice difficulty.



Figure UR-EV-10. State: Chess play entry. What is visible: Game-start choices and controls. Why it matters: Shows one route to a reviewable game. Limitation: A completed game is not shown.

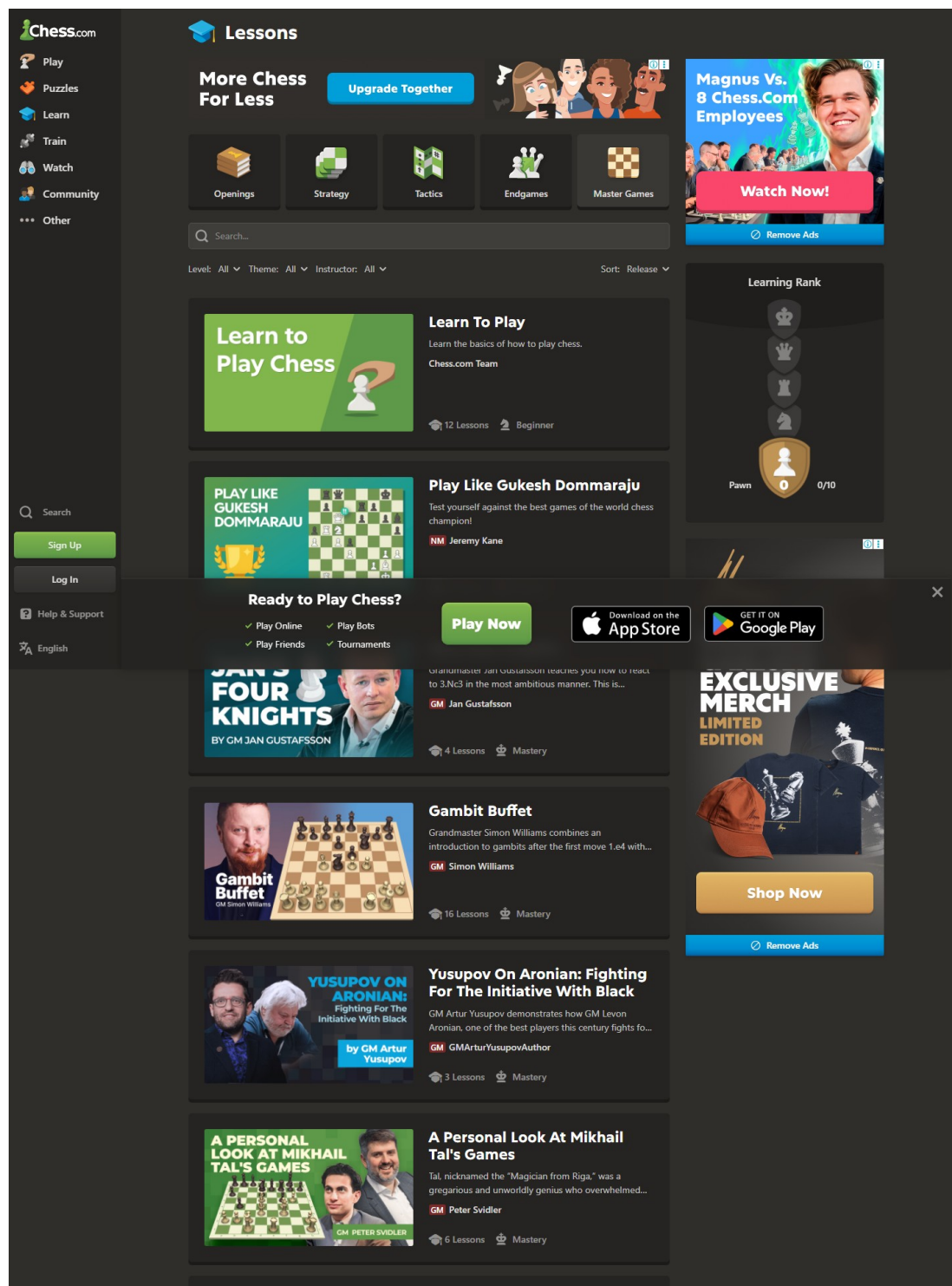


Figure UR-EV-11. State: Chess lessons. What is visible: Lesson groups and progression choices. Why it matters: Shows the learning catalog available after review. Limitation: The capture does not prove which lesson is relevant to a mistake.



Figure UR-EV-12. State: Chess puzzle entry. What is visible: Puzzle task and practice entry. Why it matters: Shows a possible continuation action. Limitation: Relevance to a specific reviewed game is not established.



Figure UR-EV-13. State: Chess analysis entry. What is visible: Board, move list, and analysis controls. Why it matters: Shows the density a beginner must interpret. Limitation: The capture does not validate a complete Game Review journey.

9. Findings

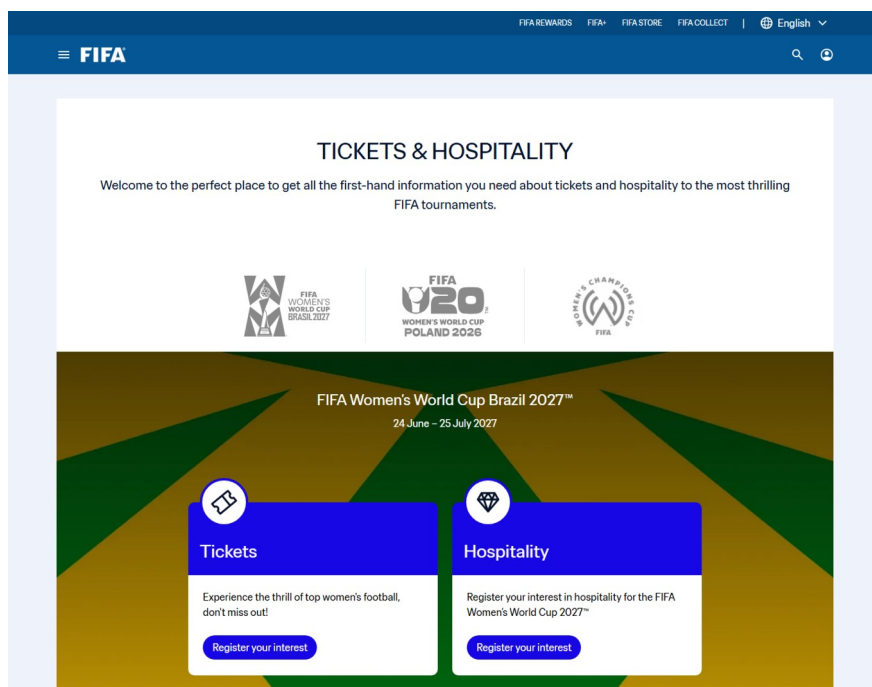


Figure UR-EV-01. FIFA ticket decision surface used by UR-F01 and UR-F02

Source: Screen evidence F2-E09; captured interface state only

Related: Ticket status clarity, freshness, handoff trust, and TP-FIFA



Figure UR-EV-02. Chess analysis-entry decision surface used by UR-C01 and UR-C02

Source: Screen evidence C2-E10; captured interface state only

Related: Entry choice overload, learning priority, vocabulary load, and TP-CHESS

9.1 Findings Summary

Finding	Evidence	Pattern	Design direction
UR-F01	F2-E09; F-SIM-04/05/06	Ticket state and freshness are separated	Show state, freshness, and next action together
UR-F02	F2-E09; F-SIM-02/05	Destination purpose and return path are unclear	Preview official domain, account, and return
UR-F03	F2-E02/E03/E04; F-SIM-03/04	Mobile browsing loses tournament context	Persist a compact tournament summary
UR-C01	C2-E10; C-SIM-01/02/05	Beginners face several review entries	Place Beginner Review beside the game result
UR-C02	C2-E07/E08/E10; C-SIM-02/04/06	Advanced detail precedes one learning priority	Show one mistake and plain-language reason first
UR-C03	C2-E05/E07/E10; C-SIM-03/04/06	Review does not lead to relevant practice	Link one lesson or puzzle to the mistake

10. Evidence boundary and citations

Screen evidence: F2-E02-E04 and F2-E09 support FIFA navigation/status claims; C2-E05, C2-E07-E10 support Chess learning and analysis-entry claims.

PA1 inherited finding: The product pair, drawbacks, and evidence limits remain continuous with PA1.

Simulated study: Twelve scenario records generate hypotheses and traceable synthetic metrics only.

Design inference: Effectiveness, efficiency, learnability, memorability, errors, satisfaction, visibility, recognition, progressive disclosure, control, and prevention guide the implications.

PA3 validation target: Prototype testing must replace synthetic expectations with observed user evidence.

Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 -

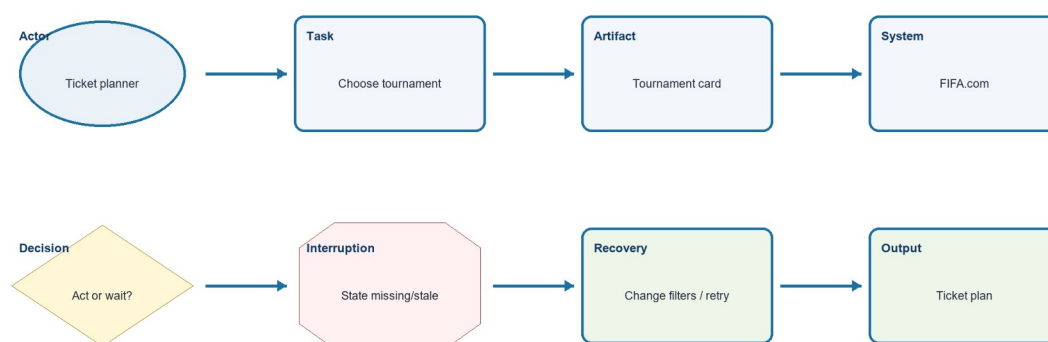
Fundamental Concepts - Usability Dimensions_2.pdf, pp. 62-67, 74-76, 83-

84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36;
LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

11. Revised Work Models

Each model uses a distinct shape for actor, task/artifact/system, decision, interruption, recovery, and output. The flow reads left to right. Captions identify the current state, problem point, user decision, recovery path, and design opportunity.

FIFA Ticket Planning Task Flow



Problem point: Status is missing or stale

Recovery path: Keep context and retry or wait

Design opportunity: Place state, freshness, and options together

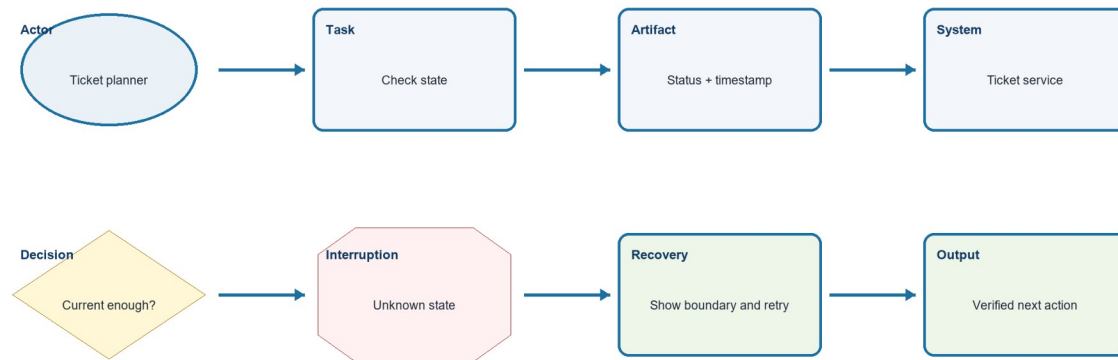
Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-FIFA-PLANNING. Wm Fifa Planning

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

FIFA Ticket Verification Sequence



Problem point: Freshness cannot be established

Recovery path: Do not infer a sale outcome

Design opportunity: Expose timestamp and source boundary

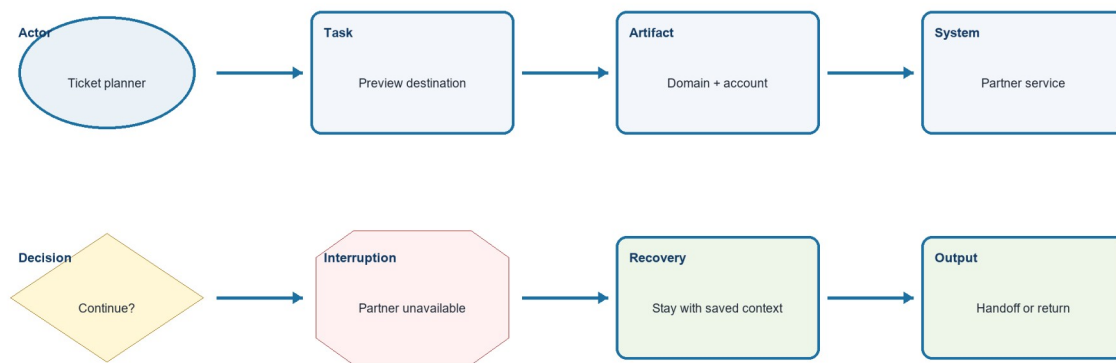
Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-FIFA-VERIFICATION. Wm Fifa Verification

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

FIFA Partner Handoff Flow



Problem point: Partner service is unavailable

Recovery path: Stay on FIFA.com with context

Design opportunity: Explain destination before departure

Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-FIFA-HANDOFF. Wm Fifa Handoff

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

FIFA Artifact and Information Model

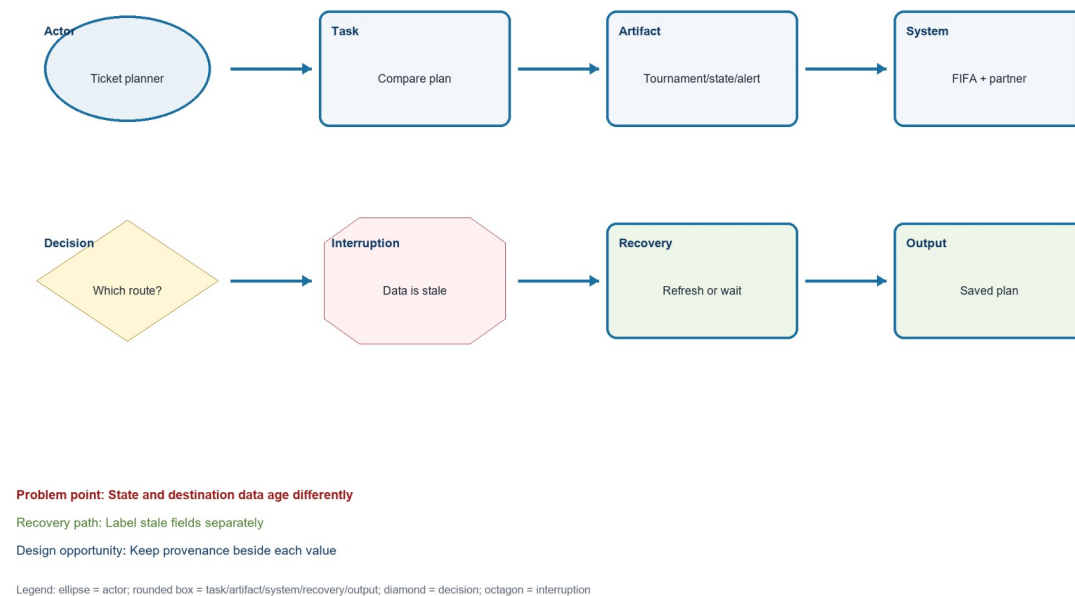


Figure UR-WM-FIFA-ARTIFACT. Wm Fifa Artifact

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

Chess Beginner Learning Flow

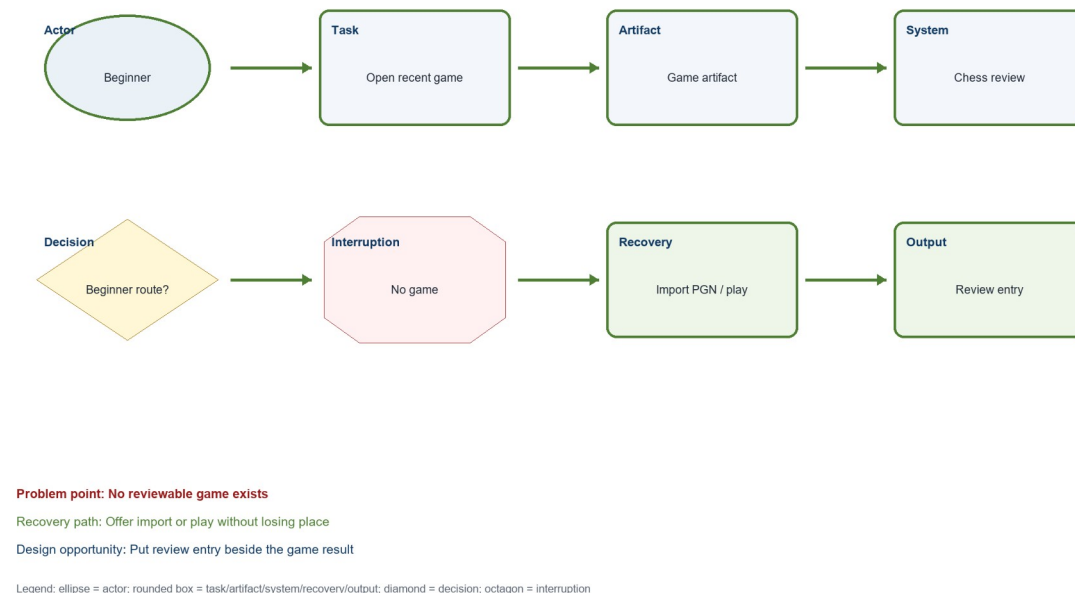
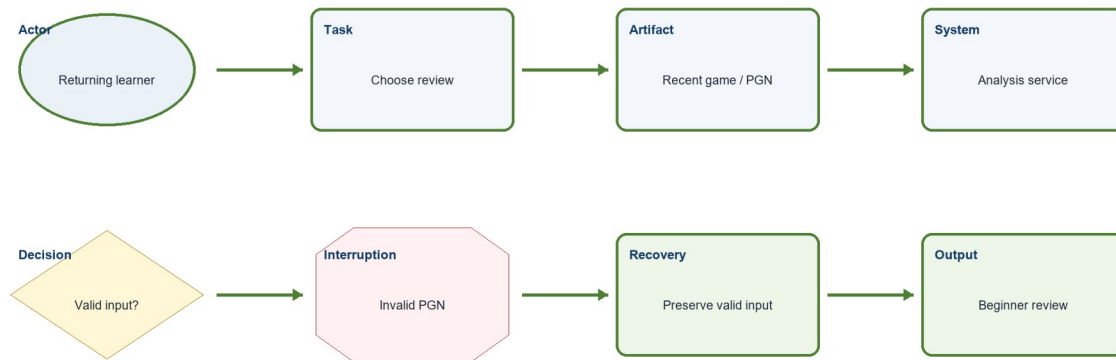


Figure UR-WM-CHESS-LEARNING. Wm Chess Learning

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

Chess Review-Entry Sequence



Problem point: PGN or game input fails validation

Recovery path: Point to the failing input and keep the rest

Design opportunity: Separate Beginner Review from advanced setup

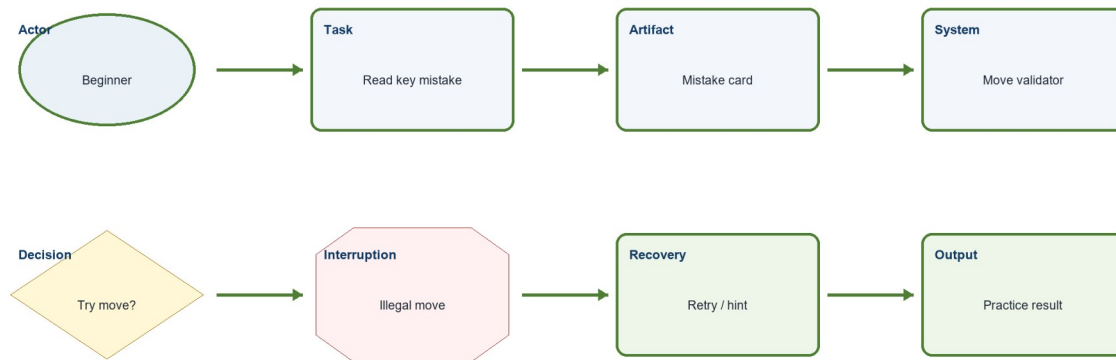
Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-CHESS-ENTRY. Wm Chess Entry

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

Chess Explanation-to-Practice Flow



Problem point: A move is illegal or service fails

Recovery path: Keep the position and allow retry

Design opportunity: Explain legality separately from quality

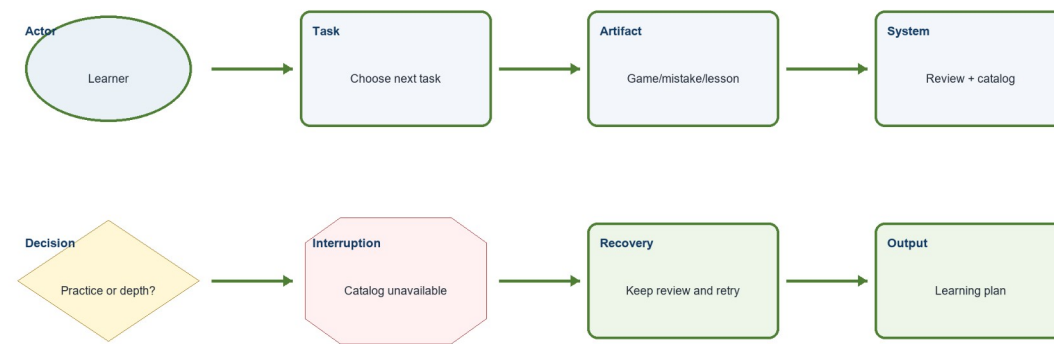
Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-CHESS-PRACTICE. Wm Chess Practice

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

Chess Learning Artifact Model



Problem point: Relevant practice cannot load

Recovery path: Preserve the review result

Design opportunity: Link one specific lesson or puzzle

Legend: ellipse = actor; rounded box = task/artifact/system/recovery/output; diamond = decision; octagon = interruption

Figure UR-WM-CHESS-ARTIFACT. Wm Chess Artifact

Source: PA2 design QA artifact

Related: FIFA or Chess task evidence, interruption, recovery, and design opportunity

Work Model Coverage Checklist

Model	Trigger	System response	Decision / interruption	Recovery	Output / opportunity
FIFA ticket planning	Open ticket planning	Load tournament and route state	Act now or wait; state may be missing	Change filters or retry	Ticket plan; combine state and action
FIFA ticket verification	Inspect freshness	Return timestamp and source	Current enough; status may be unknown	Keep last labeled update and retry	Verified next action; expose provenance
FIFA partner handoff	Choose an outbound route	Preview domain and account boundary	Continue or stay; partner may fail	Stay with saved context	Handoff or return; explain destination
FIFA information model	Compare saved plan	Assemble event, state, alert, and source	Choose a route; data may be stale	Refresh only stale fields	Saved plan; keep provenance beside

Model	Trigger	System response	Decision / interruption	Recovery	Output / opportunity
					values
Chess beginner learning	Open a recent game	Offer Beginner Review	Choose route; no game may exist	Import PGN or play	Review entry; place action beside result
Chess review entry	Select review	Validate recent game or PGN	Valid input; PGN may fail	Correct input without losing metadata	Guided review; identify failing line
Chess explanation to practice	Open key mistake	Show reason and one practice link	Continue or reveal depth; service may fail	Retry or keep board-only view	Practice action; explain relevance
Chess learning artifact	Inspect review artifact	Link position, reason, lesson, and depth	Choose next artifact; content may be unavailable	Return to saved game	Learning path; preserve beginner focus

12. Practical Difficulties and Lessons Learned

Practical difficulty	How the group handled it	Lesson learned	Impact on the final version
Screenshots did not always show the state needed for a claim.	The group separated captured state from simulated behavior and design inference.	A screenshot proves only the visible state and capture time.	Claims now name their evidence boundary and recovery assumptions.
Desktop and mobile pages expose different ordering and page length.	The group used separate mobile evidence and task models.	Cross-device continuity must be specified as a requirement.	The revised models preserve tournament/game context during recovery.

Appendix A - Twelve Detailed Simulated Note Sheets

A.1 Note sheet F-SIM-01

Simulated observation: F-SES-01 - 16 Jul 2026 - 28 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-01; Casual football fan; 18-24; Android phone; motivation: Find ticket entry without planning expertise; constraint: Limited time and data; accessibility: Large tap targets
Task	F1, F2, F4, F5
Action sequence	Opens menu, scrolls to Tickets & Hospitality, returns to Match Centre, then reopens ticket page.
Hesitation point	Pauses at tournament cards with different CTA wording.
Visible interface trigger	Tickets & Hospitality card and tournament status text.
Inferred mental model	Ticket page should give one current action.
Simulated quote	"I can find tickets, but I still do not know what is open."
Simulated task result	Partial success
Workaround	Uses Match Centre as a comparison anchor.
Design opportunity	Show state, freshness, and next action together.
Limitation	Scenario-based; no live sale data.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-01 without hiding needed control.

A.2 Note sheet F-SIM-02

Simulated observation: F-SES-02 - 16 Jul 2026 - 31 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-02; Casual football fan; 25-34; Windows laptop + phone; motivation: Compare match and ticket status; constraint: Avoids creating accounts early; accessibility: High contrast
Task	F1-F6
Action sequence	Finds Match Centre quickly, compares schedule, opens ticket entry, then checks destination preview.
Hesitation point	Hesitates before the external handoff.
Visible interface trigger	Match Centre filters and partner CTA.
Inferred mental model	An official domain preview should precede departure.
Simulated quote	"The schedule is clear; the handoff is the risky part."
Simulated task result	Success with hesitation
Workaround	Opens destination in a second tab.
Design opportunity	Keep tournament context and return route.
Limitation	Scenario-based; partner response not tested.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-02 without hiding needed control.

A.3 Note sheet F-SIM-03

Simulated observation: F-SES-03 - 17 Jul 2026 - 30 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-03; Tournament follower; 25-34; iPhone; motivation: Coordinate a group trip; constraint: Needs fast resumption; accessibility: Text zoom
Task	F1-F6
Action sequence	Selects tournament, saves context mentally, moves to phone view, and searches for the same state.
Hesitation point	Loses orientation after the mobile stack changes order.
Visible interface trigger	Long mobile tournament-card stack.
Inferred mental model	Mobile should restore the selected tournament first.
Simulated quote	"I want the same tournament to stay at the top."
Simulated task result	Partial success
Workaround	Uses page search and browser history.
Design opportunity	Persist context across devices.
Limitation	Scenario-based; account synchronization assumed.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-03 without hiding needed control.

A.4 Note sheet F-SIM-04

Simulated observation: F-SES-04 - 17 Jul 2026 - 34 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-04; Tournament follower; 35-44; MacBook + Android phone; motivation: Connect schedule context to ticket action; constraint: Many open tabs; accessibility: Keyboard access on laptop
Task	F1-F6
Action sequence	Uses Match Centre filters, opens ticket card, checks freshness, then returns.
Hesitation point	Questions whether the status changed since the last visit.
Visible interface trigger	Status label without a prominent timestamp.
Inferred mental model	Current state requires visible provenance.
Simulated quote	"Is this from today or from the last sale window?"
Simulated task result	Failure on freshness
Workaround	Refreshes and compares wording.
Design opportunity	Display updated time and governed source.
Limitation	Scenario-based; timestamp policy unknown.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-04 without hiding needed control.

A.5 Note sheet F-SIM-05

Simulated observation: F-SES-05 - 18 Jul 2026 - 33 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-05; Trust-sensitive ticket planner; 35-44; Windows laptop; motivation: Avoid unofficial or stale destinations; constraint: Needs provenance before leaving site; accessibility: Clear status text, not color alone
Task	F1-F6
Action sequence	Compares buy, wait, resale, and hospitality paths before choosing to stay.
Hesitation point	Stops when official destination is not explained.
Visible interface trigger	Multiple commercial paths and outbound CTA.
Inferred mental model	Departure requires trust evidence and safe cancellation.
Simulated quote	"I will not leave until I know where this goes."
Simulated task result	Success by choosing stay
Workaround	Copies the domain name for separate checking.
Design opportunity	Preview destination, purpose, and return behavior.
Limitation	Scenario-based; no transaction attempted.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-05 without hiding needed control.

A.6 Note sheet F-SIM-06

Simulated observation: F-SES-06 - 18 Jul 2026 - 35 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant F-SIM-06; Trust-sensitive ticket planner; 45-54; Android phone; motivation: Know whether to buy, wait, or register interest; constraint: Low confidence after handoff; accessibility: Plain language and persistent focus
Task	F1-F6
Action sequence	Finds ticket page, encounters missing state, retries on variable network, then subscribes conceptually.
Hesitation point	Cannot distinguish missing from sold out.
Visible interface trigger	Empty or stale status region.
Inferred mental model	Missing data is not a ticket outcome.
Simulated quote	"No status should not look like no tickets."
Simulated task result	Recovered after retry
Workaround	Returns to homepage and tries again.
Design opportunity	Differentiate missing, stale, sold out, and waiting room.
Limitation	Scenario-based; notification permission simulated.

PA3 validation target: Test whether the proposed response reduces the hesitation in F-SES-06 without hiding needed control.

A.7 Note sheet C-SIM-01

Simulated observation: C-SES-01 - 16 Jul 2026 - 29 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-01; First-time beginner; 18-24; Android phone; motivation: Choose a safe beginner route; constraint: Unfamiliar analysis terms; accessibility: Plain language
Task	C1-C3, C7-C9
Action sequence	Chooses Learn, continues a lesson, solves a puzzle, then looks for review.
Hesitation point	Opens several analysis entry choices.
Visible interface trigger	Learn cards, puzzle board, analysis entry menu.
Inferred mental model	Review should start with a beginner preset.
Simulated quote	"Which button tells me what I did wrong?"
Simulated task result	Partial success
Workaround	Returns to the completed game screen.
Design opportunity	Offer one labeled beginner route.
Limitation	Scenario-based; no engine output claimed.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-01 without hiding needed control.

Simulated observation: C-SES-02 - 16 Jul 2026 - 32 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-02; First-time beginner; 25-34; Windows laptop; motivation: Understand one mistake; constraint: Fears breaking the position; accessibility: Keyboard-only path
Task	C1-C9
Action sequence	Uses keyboard to open a game, selects beginner review, tries a move, and reads help.
Hesitation point	Focus order reaches advanced controls before explanation.
Visible interface trigger	Analysis controls and terminology links.
Inferred mental model	The main mistake should precede advanced depth.
Simulated quote	“I found analysis, but not the first thing to learn.”
Simulated task result	Success with wrong path
Workaround	Uses Tab and headings to relocate.
Design opportunity	Order content by learning priority.
Limitation	Scenario-based; keyboard path modeled.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-02 without hiding needed control.

A.9 Note sheet C-SIM-03

Simulated observation: C-SES-03 - 17 Jul 2026 - 27 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-03; Returning learner; 18-24; iPhone; motivation: Resume without relearning navigation; constraint: Forgets feature locations; accessibility: Text zoom
Task	C1-C3, C5-C9
Action sequence	Recognizes puzzles, resumes a lesson, opens a completed game, then searches for analysis.
Hesitation point	Forgets the name of the review feature.
Visible interface trigger	Navigation labels and game actions.
Inferred mental model	Returning learners recognize outcomes, not feature terms.
Simulated quote	"I remember the result, not the menu name."
Simulated task result	Recovered
Workaround	Uses recent activity to reopen the game.
Design opportunity	Add review action at the completed game.
Limitation	Scenario-based; recency history assumed.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-03 without hiding needed control.

A.10 Note sheet C-SIM-04

Simulated observation: C-SES-04 - 17 Jul 2026 - 35 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-04; Returning learner; 35-44; MacBook; motivation: Reconnect review to practice; constraint: Forgets analysis vocabulary; accessibility: Reduced motion
Task	C1-C9
Action sequence	Opens game, selects review, reads one explanation, then continues to a lesson.
Hesitation point	Stops at evaluation and engine terms.
Visible interface trigger	Evaluation label and advanced analysis controls.
Inferred mental model	Terminology help should be available in place.
Simulated quote	"Tell me why before showing the numbers."
Simulated task result	Success with hesitation
Workaround	Opens help text and ignores depth.
Design opportunity	Plain-language explanation with optional terms.
Limitation	Scenario-based; explanation quality not measured.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-04 without hiding needed control.

Simulated observation: C-SES-05 - 18 Jul 2026 - 30 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-05; Intermediate player with low analysis use; 25-34; Windows laptop; motivation: Review without engine overload; constraint: Skips dense panels; accessibility: Keyboard shortcuts with labels
Task	C1-C9
Action sequence	Finds analysis, follows one candidate line, returns, and selects a puzzle.
Hesitation point	Takes a wrong path into full analysis first.
Visible interface trigger	Multiple setup paths and engine panel.
Inferred mental model	Low-analysis users need a preset at entry.
Simulated quote	“I opened the expert view by accident.”
Simulated task result	Recovered
Workaround	Backs out and uses review link.
Design opportunity	Default to beginner review for the scenario.
Limitation	Scenario-based; recommendation accuracy not claimed.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-05 without hiding needed control.

Simulated observation: C-SES-06 - 18 Jul 2026 - 34 min; structured notes only; no audio/video.

Field	Record
Participant profile	Simulated participant C-SIM-06; Intermediate player with low analysis use; 35-44; Android tablet; motivation: Find one actionable improvement; constraint: Terminology slows scanning; accessibility: Color-independent indicators
Task	C1-C9
Action sequence	Opens recent game, identifies one turning point, tries a move, and reveals advanced detail.
Hesitation point	Scans slowly on tablet because labels compete.
Visible interface trigger	Board, move list, explanation, depth disclosure.
Inferred mental model	A single learning card should lead the visual hierarchy.
Simulated quote	“One mistake and one next exercise is enough.”
Simulated task result	Success
Workaround	Collapses side panels.
Design opportunity	Progressive disclosure and continuation action.
Limitation	Scenario-based; engine availability assumed.

PA3 validation target: Test whether the proposed response reduces the hesitation in C-SES-06 without hiding needed control.